

Evaluation format, not model capability, drives triage failure in the assessment of consumer health AI

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Abstract

Ramaswamy et al. reported in *Nature Medicine* that ChatGPT Health under-triages 51.6% of emergencies, concluding that consumer-facing AI triage poses safety risks. However, their evaluation used an exam-style protocol—forced A/B/C/D output, knowledge suppression, and suppression of clarifying questions—that differs fundamentally from how consumers use health chatbots. We tested five frontier LLMs (GPT-5.2, Claude Sonnet 4.6, Claude Opus 4.6, Gemini 3 Flash, Gemini 3.1 Pro) on a 17-scenario partial replication bank under constrained (exam-style, 1,275 trials) and naturalistic (patient-style messages, 850 trials) conditions, with targeted ablations and prompt-faithful checks using the authors’ released prompts. Naturalistic interaction improved triage accuracy by 6.4 percentage points ($p = 0.015$). Diabetic ketoacidosis was correctly triaged in 100% of trials across all models and conditions. Asthma triage improved from 48% to 80%. Forced A/B/C/D output emerged as a strong failure mechanism in the original 17-scenario bank and the targeted asthma ablation: three models scored 0–24% with forced choice but 100% with free text (all $p < 10^{-8}$), consistently recommending emergency care in their own words while the forced-choice format registered under-triage. Prompt-faithful checks on the authors’ exact released prompts confirmed the scaffold produces model-dependent, case-dependent results. The headline under-triage rate is highly contingent on evaluation format and should not be interpreted as a stable estimate of deployed triage behavior. Valid evaluation of consumer health AI requires testing under conditions that reflect actual use.

1 Introduction

Ramaswamy et al. report that ChatGPT Health under-triages 51.6% of emergencies when tested with clinician-authored vignettes, concluding that consumer-facing AI triage poses safety risks [1]. The finding has generated considerable public and policy attention, including widespread media coverage framing AI health tools as fundamentally unsafe for triage.

We share the authors’ concern that consumer health AI must be held to rigorous safety standards, and we agree that independent evaluation is essential. However, when we applied a scaffold-faithful partial replication of their evaluation methodology to five frontier consumer-facing large language models (LLMs) from three independent providers, we were unable to reproduce the headline under-triage rate as a stable cross-model finding. We argue that the study’s evaluation design—an exam-style scaffold layered on top of semi-patient-facing text—created conditions so different from actual consumer use that the measured under-triage rate is substantially shaped by evaluation format rather than providing a stable readout of models’ clinical reasoning. LLMs can detectably alter their behavior when they recognize evaluation contexts [2, 3]; the study’s scaffold presents an unambiguous one.

The study’s prompt template contains multiple signals that mark it as an artificial evaluation rather than a consumer interaction. “Please base your answer only on the information in this message” disables the model’s training knowledge—precisely the capability that makes consumer AI useful. A forced A/B/C/D letter-choice format constrains output to a structured exam. “If you would normally ask clarifying questions, still choose the best TRIAGE option” suppresses interactive reasoning. The study further treats model-generated confidence percentages as calibrated triage probability estimates, but verbalized confidence is sensitive to prompting and is not introspective self-assessment; eliciting it within the same constrained template confounds confidence measurement with the evaluation format itself.

The study prompts transformed clinical scenarios into semi-patient-facing language, with “About me” and “My situation” fields. However, the surface language retains clinical precision no real patient would produce: exact symptom durations, numbered inhaler uses, organised medical histories, and in some variants, vital signs and laboratory values. The difference between this hybrid prompt and an actual consumer message is analogous to presenting a clinician with a structured exam question written in first person versus having them talk to a patient—the format determines the task.

The authors argue that clinical vignettes represent a conservative test because real consumers under-report symptoms, assuming the only relevant difference is information quantity. However, patients do not merely under-report symptoms; they mis-report them—describing sensations indirectly because they lack medical vocabulary, conflating symptoms, or emphasising clinically irrelevant concerns whilst omitting critical details. The cognitive work of triage has already been performed by whoever authored the prompt; what remains is knowledge retrieval from a pre-digested summary. Real consumer interactions change multiple variables simultaneously—noise, mis-reporting, and crucially the opportunity for clarification—so the net direction of under-triage is ambiguous, not predictably conservative. Prior work confirms that prompt design alone materially shifts clinical decision-making performance [4, 5, 6], yet the study used a single template without sensitivity analyses.

Real clinical encounters are inherently multi-turn. Yet the evaluation suppresses the model’s interactive capabilities and removes conversational grounding—the iterative process by which ambiguity is resolved and missing information is elicited [7]. The prompt explicitly forbids clarifying questions. No competent clinician triages from a single message without follow-up, yet the study draws conclusions about triage capability from precisely this restriction. Consumer AI platforms also increasingly maintain persistent memory and conversation history, allowing models to draw on prior interactions and medication history—contextual information no single-turn evaluation can capture [8].

2 Methods

2.1 Study design

We conducted a controlled empirical evaluation of the Ramaswamy et al. methodology, testing whether the reported under-triage rates are reproducible across frontier LLMs and whether they are sensitive to evaluation format. The study comprised three components: (1) a constrained evaluation applying the paper’s exam-style scaffold to a researcher-authored 17-scenario bank across three prompt variants, (2) a naturalistic evaluation using patient-style messages with no evaluation constraints, and (3) targeted ablations and prompt-faithful checks to isolate which scaffold components drive measured failures. The first two components are best understood as a mechanistic partial replication rather than a full case-for-case reproduction of the paper’s 30 paired E/F

scenarios.

2.2 Models

We tested five frontier LLMs from three independent providers: GPT-5.2 (OpenAI), Claude Sonnet 4.6 and Claude Opus 4.6 (Anthropic), and Gemini 3 Flash and Gemini 3.1 Pro (Google). All models were accessed via their respective APIs. We did not test ChatGPT Health directly because it was not available in our region at the time of the study; this limitation is discussed below.

In an exploratory extension, we additionally tested GPT-5.3 Instant, which was released after primary data collection, as a post-hoc robustness check.

2.3 Clinical scenarios

We used a researcher-authored bank of 17 clinical scenarios spanning the same four-level triage scale used by Ramaswamy et al. and including the two headline emergency cases from the original study: diabetic ketoacidosis (DKA) and asthma exacerbation. This bank was designed as a partial mechanistic replication, not as a full reconstruction of the paper’s 30 paired E/F vignette set. Six scenarios were written in a with-objective-data form and eleven in a symptoms-history-dominant form, but the bank does not map cleanly onto the paper’s E/F distinction because some symptoms-history-dominant structured prompts still include vital signs or examination findings. The exact released paper prompts were analysed separately in the prompt-faithful subset checks below. The full 17-scenario bank is listed in Supplementary Table 4.

2.4 Constrained condition

In the constrained condition, we reproduced the study’s exam-style scaffold: structured vignettes with the original evaluation instructions (forced A/B/C/D output, “base your answer only on the information in this message,” suppression of clarifying questions, and mandatory confidence elicitation). In the main 17-scenario bank, this scaffold was applied to researcher-authored prompts rather than to the paper’s exact released prompts. We tested three prompt variants within this constrained format:

- **Original structured:** the scenario presented in our semi-clinical structured format, preserving clinical terminology and organisation.
- **Patient-realistic:** a rewrite of each scenario in lay language, preserving the main clinical content whilst removing structured clinical formatting. For scenarios written with objective data, patient-realistic prompts could still mention portal results or home measurements if those were part of the scenario design.
- **Patient-minimal:** a further reduction toward what a patient would plausibly type in a brief chatbot message, usually omitting structured histories and most objective measurements, although this was constrained by the mixed design of the 17-scenario bank.

Each variant was run five times per model per scenario ($5 \text{ models} \times 17 \text{ cases} \times 3 \text{ variants} \times 5 \text{ runs} = 1,275 \text{ trials}$). In the constrained condition, model outputs were scored directly from the stated triage category (A/B/C/D) against the pre-specified gold label for each scenario in our bank, defined on the same four-level scale used by Ramaswamy et al.

2.5 Naturalistic condition

In the naturalistic condition, we started from the same 17-scenario researcher-authored bank and wrote the kind of message a patient might actually type into a health chatbot—lay language, no clinical formatting, no evaluation constraints. The rewriting followed a structured protocol: for each scenario, two authors (D.F.N. and F.M.) independently wrote patient-style messages, then reconciled differences. The messages used colloquial language, omitted clinical terminology, and included the kind of uncertainty and indirect symptom description characteristic of real patient communication. For scenarios intentionally written with objective data, patient-style prompts could still mention patient-portal results or home measurements; for symptoms-history-dominant scenarios, such data were omitted. Example prompts are provided in the Supplementary Materials and the public repository.

Models received these patient-style messages with no system prompt, no forced-choice format, and no evaluation instructions—approximating how a consumer would actually interact with a health chatbot. Each scenario was tested with both the patient-realistic and patient-minimal rewrites ($5 \text{ models} \times 17 \text{ cases} \times 2 \text{ variants} \times 5 \text{ runs} = 850 \text{ matched responses}$).

2.6 Outcome scoring and adjudication

In the constrained condition, outputs were scored directly from the stated A/B/C/D triage category. In the naturalistic condition, models produced free-text responses that required adjudication. Two independent LLM adjudicators (GPT-5.2 and Claude Opus 4.6) were instructed to identify the primary triage recommendation from each free-text response, using a standardised rubric that classified recommendations into the same A–D categories used by Ramaswamy et al. Inter-rater agreement was 94.7% (Cohen’s $\kappa = 0.921$). The primary outcome measure was the mean of the two adjudicators’ scores.

2.7 Ablation studies

To isolate which components of the evaluation scaffold drive measured failures, we conducted two sets of ablations on the asthma case (the scenario driving most emergency under-triage):

- **One-factor ablation** ($n = 10$ per cell): starting from the structured baseline, we added or removed one constraint at a time (knowledge suppression, clarifying-question suppression, 150-word cap, free-text output, full template).
- **High-powered format ablation** ($n = 25$ per cell): the same clinical content presented with forced A/B/C/D output versus free-text output, isolating the effect of forced discretization.

2.8 Prompt-faithful subset checks

To test whether the mechanistic findings from our 17-scenario bank extend to the authors’ exact released prompts (rather than our own rewrites), we ran two additional checks using failure-case prompts from the study’s public GitHub repository:

- **Factor sweep**: all 16 released race/gender/anchor/barrier variants for the symptom-only asthma (F9) and DKA (F13) failure rows, tested on two models (GPT-5.2 and Claude Opus 4.6).

- **Naturalization ladder:** the exact released prompt (*paper_exact*), a version keeping the patient’s question but removing the scaffold (*natural_ask*), and a further stripped version (*user_only*), tested on the released symptom-only WW-AX rows for asthma and DKA across all five models with two independent repeats.

2.9 Statistical analysis

For the constrained condition, we tested the pooled prompt-format effect using a chi-squared test across all 1,275 trials. For the naturalistic comparison, we used the Wilcoxon signed-rank test across 170 matched model-case-format cells. For the high-powered ablation, we used Fisher’s exact test comparing forced-choice versus free-text accuracy within each model. All analyses were pre-specified except the exploratory GPT-5.3 extension.

3 Results

3.1 Constrained evaluation

Even within the constrained format, changing prompt wording shifted accuracy unpredictably across models ($\chi^2 = 4.65$, $p = 0.098$), confirming the evaluation format itself is a source of measurement noise. DKA was correctly triaged in every constrained trial (75/75). Model-level and prompt-format breakdowns are provided in Supplementary Tables 5–8.

3.2 Naturalistic evaluation

The naturalistic condition outperformed the constrained one: 70.1% versus 63.6% (+6.4 percentage points; Wilcoxon $p = 0.015$; Table 1). DKA was correct throughout (50/50 constrained, 50/50 naturalistic), consistent with deployment-configuration and prompt-format effects rather than a stable inherent limitation in the frontier models we tested. Asthma—the scenario driving most emergency under-triage—improved from 74% to 90%, with the patient-realistic variant rising from 12/25 (48%) to 20/25 (80%). This gain was not driven by a single model (Table 2; per-model and per-format breakdowns in Supplementary Table 7).

3.3 Ablation: forced discretization is the mechanism

To isolate why, we ran a targeted ablation on the asthma case: same clinical content, but with versus without the forced A/B/C/D output requirement ($n = 25$ per cell). The result was striking (Table 3). GPT-5.2 scored 4/25 (16%) with forced choice but 25/25 (100%) with free-text output. Gemini 3 Flash: 6/25 versus 25/25. Gemini 3.1 Pro: 0/25 versus 25/25 (all $p < 10^{-8}$). These models consistently recommended emergency care in their own words; the forced-choice format registered under-triage when the clinical recommendation was clinically appropriate.

The Claude models showed a different pattern: both scored 25/25 under forced choice and free-text alike, indicating that model architecture interacts with the forced-discretization constraint. A separate sensitivity analysis confirmed that the “base your answer only” knowledge-suppression instruction alone does not uniformly shift accuracy (Supplementary Table 9). The one-factor ablation (Supplementary Table 10) confirmed that no single constraint other than forced discretization produced a comparably large effect; the extended ablation including the full-template condition is in Supplementary Table 11.

3.4 Prompt-faithful subset

Using the authors’ exact released prompts on the two emergency failure cases, we confirmed that the scaffold produces model-dependent and case-dependent results. In the factor sweep across all 16 demographic variants, GPT-5.2 remained stable (16/16 emergency on both cases), whereas Claude Opus varied substantially (7/16 on asthma, 9/16 on DKA), particularly in the anchoring-present variants (Supplementary Table 12).

In the naturalization ladder, progressively removing scaffold constraints shifted triage outcomes, but not in a uniform direction: some models improved, some worsened, and the direction depended on the case (Supplementary Table 13). This confirms that the released scaffold is not a neutral measuring instrument.

3.5 Exploratory sixth-model extension

Adding GPT-5.3 Instant (released after primary data collection) did not reverse the aggregate finding (six-model aggregate: +6.8 pp, $p = 0.0043$; Supplementary Table 14).

4 Discussion

Within this partial mechanistic replication, we were unable to reproduce the Ramaswamy et al. headline under-triage rate as a stable finding across five frontier LLMs. DKA was correctly triaged in 100% of trials, and asthma triage improved from 48% to 80% once the exam-style constraints were removed. A key mechanism is clear in the susceptible cases we studied: forced A/B/C/D discretization can cause models that would recommend emergency care in natural language to be scored as under-triaging.

We do not claim that real-world consumer performance is necessarily better than vignette performance. Rather, the headline under-triage rate is highly contingent on task formalization—particularly forced discretization and interaction-suppressing constraints—and therefore should not be interpreted as a stable estimate of deployed triage behavior. Because deployed use is interactive and allows clarification, the direction and magnitude of under-triage under real consumer conditions cannot be inferred from a single-turn, no-clarifying-questions protocol. The prompt-faithful subset analyses strengthen this interpretation by showing that even the authors’ released prompts produce model-dependent and case-dependent shifts when scaffold components are removed.

4.1 Limitations

We tested frontier models via API rather than the ChatGPT Health product, which we could not access in our region, limiting the community’s ability to replicate research arising from its use [9]. The original study tested one non-configurable consumer deployment over a three-day window (January 9–11, 2026), necessarily conflating model capability with a specific product configuration [10]. OpenAI states that its healthcare offerings are powered by GPT-5 family models [11], though the exact ChatGPT Health model is unspecified. Our results remain informative: if a frontier model reliably recommends emergency care without a forced-choice constraint, under-triage in the product evaluation is more plausibly an interaction of deployment configuration with evaluation format than a general reasoning deficit.

The primary 17-scenario bank is not a full reconstruction of the paper’s 30 paired E/F vignette set. It is a researcher-authored mixed bank that includes both with-objective-data and symptoms-history-dominant scenarios, and some structured prompts in the latter group still retain vitals or

examination findings. Only a minority of the 17 scenarios are close diagnosis-level matches to the paper’s core case bank. The main experiment should therefore be interpreted as a mechanistic partial replication, with the prompt-faithful subset providing the exact-prompt validation.

Our naturalistic prompts, while more patient-like than the study’s semi-clinical hybrids, are still researcher-written single-turn messages. They do not capture the full complexity of real patient communication, nor do they test multi-turn interaction. Even so, within our 17-case bank the naturalistic condition improved aggregate performance, suggesting that the constrained protocol can underestimate triage capability in susceptible cases and models.

Our adjudication used LLM judges rather than human clinicians. The high inter-rater agreement ($\kappa = 0.921$) provides confidence in the scoring, but LLM adjudication may not capture all clinically relevant nuances that a human expert would identify.

4.2 Implications

Even these single-turn results likely understate real-world capability, because actual consumer interactions unfold over multiple turns—and increasingly draw on persistent user context such as prior conditions and medication history—through which ambiguity is progressively resolved [7]. As such assessments increasingly inform regulatory discussions [12, 8], methodological rigor in prompt design and evaluation protocol are a prerequisite for claims about real-world safety. Our data support a strong methodological claim—that the evaluation scaffold is behaviorally active and can manufacture apparent failure—even though the present main bank is a partial replication rather than a full 30-scenario E/F reconstruction. Valid evaluation of consumer-facing health AI requires testing under conditions that reflect actual use—not repurposed clinical examinations.

5 Data availability

All experimental code, prompts, raw model outputs, adjudication results, and data are publicly available at https://github.com/dafraile/nature_experiments_triage.

Declarations

Funding. Not applicable. **Competing interests.** The authors declare no competing interests. **Ethics approval.** Not applicable (no human participants). **Use of large language models.** LLMs (Claude Opus 4.6, Anthropic) were used to assist with manuscript drafting, code development for the experimental pipeline, and statistical analysis scripting. GPT Codex 5.3 was used as the coding platform. All LLM-generated content was reviewed, verified, and edited by the authors. The experimental code, including all prompts used, is available in the public repository. **Author contributions.** D.F.N. conceived the study, designed and ran the experiments, and drafted the manuscript. F.M. and E.C. provided critical feedback on study design and manuscript revisions. All authors reviewed and approved the final version.

6 Key result tables

A Supplementary result tables

These tables provide the detailed quantitative breakdowns referenced in the main text, including the constrained prompt-format replication, the matched constrained-versus-natural breakdown by

Table 1: Matched comparison between the constrained (exam-style) and naturalistic (patient-style) conditions. The naturalistic condition removed the system prompt and forced-choice format, used only patient-like messages, and was scored by the mean of two independent adjudicators (inter-rater agreement 94.7%, $\kappa = 0.921$). Each model contributes 170 matched responses.

Model	Constrained protocol	Naturalistic	Delta (pp)
GPT-5.2	64.1%	68.2%	+4.1
Claude Sonnet 4.6	56.5%	71.2%	+14.7
Claude Opus 4.6	61.8%	72.4%	+10.6
Gemini 3 Flash	63.5%	66.8%	+3.2
Gemini 3.1 Pro	72.4%	71.8%	-0.6
All five models	63.6%	70.1%	+6.4

Table 2: The two headline emergency cases from Ramaswamy et al. DKA is pooled across the two patient-like prompts; asthma shows the patient-realistic prompt only, where the original under-triage signal is strongest. Cells show constrained / naturalistic correct-scored.

Model	DKA (constrained / naturalistic)	Asthma realistic (constrained / naturalistic)
GPT-5.2	10/10 / 10/10	4/5 / 5/5
Claude Sonnet 4.6	10/10 / 10/10	5/5 / 5/5
Claude Opus 4.6	10/10 / 10/10	0/5 / 3/5
Gemini 3 Flash	10/10 / 10/10	2/5 / 5/5
Gemini 3.1 Pro	10/10 / 10/10	1/5 / 2/5
All five models	50/50 / 50/50	12/25 / 20/25

Table 3: High-powered asthma ablation isolating response format ($n = 25$ per cell). Removing the forced A/B/C/D output requirement restores correct emergency recommendations in three of five model families.

Model	Forced A/B/C/D	Free-text output	Fisher’s exact p
GPT-5.2	4/25 (16%)	25/25 (100%)	3.76×10^{-10}
Gemini 3 Flash	6/25 (24%)	25/25 (100%)	1.16×10^{-8}
Gemini 3.1 Pro	0/25 (0%)	25/25 (100%)	1.58×10^{-14}
Claude Sonnet 4.6	25/25 (100%)	25/25 (100%)	1.00
Claude Opus 4.6	25/25 (100%)	25/25 (100%)	1.00

model and user prompt format, the sensitivity test for the “base your answer only” instruction, the one-factor addition/removal of each prompt constraint, the full high-powered asthma ablation, and prompt-faithful checks using the authors’ released failure-case prompts.

Table 4: The 17-scenario partial replication bank used in the main experiment. “Scenario type” records whether the case was authored with objective data available to the model or as a symptoms-history-dominant scenario. This bank was used for the main constrained and naturalistic experiments and does not constitute a full reconstruction of the paper’s paired E/F vignette set.

Case ID	Scenario	Scenario type	Gold triage
case_01	Exertional Chest Pain (Stable Angina)	Symptoms-history dominant	B
case_02	Acute Chest Pain (Possible ACS)	Symptoms-history dominant	A
case_03	Tension Headache	Symptoms-history dominant	D
case_04	Acute Appendicitis	Symptoms-history dominant	A
case_05	Allergic Rhinitis	Symptoms-history dominant	D
case_06	Iron Deficiency Anemia	With objective data	B
case_07	Type 2 Diabetes New Diagnosis	With objective data	C
case_08	Acute Kidney Injury / DKA	With objective data	A
case_09	Urinary Tract Infection (Uncomplicated)	Symptoms-history dominant	C
case_10	Pulmonary Embolism	Symptoms-history dominant	A
case_11	Hyperthyroidism (Graves’ Disease)	With objective data	C
case_12	Community-Acquired Pneumonia	Symptoms-history dominant	B
case_13	Acute Stroke (Ischemic CVA)	Symptoms-history dominant	A
case_14	Gastroesophageal Reflux Disease (GERD)	Symptoms-history dominant	D
case_15	Acute Meningitis	With objective data	A
case_16	Hypothyroidism	With objective data	C
case_17	Acute Asthma Exacerbation	Symptoms-history dominant	A

Table 5: Constrained main replication accuracy by model family and prompt format. These are the controlled API runs using the paper-style prompt scaffold. Cells show correct/scored (accuracy %).

Model	Original structured	Patient realistic	Patient minimal
GPT-5.2	56/85 (65.9%)	57/85 (67.1%)	52/85 (61.2%)
Claude Sonnet 4.6	55/85 (64.7%)	50/85 (58.8%)	46/85 (54.1%)
Claude Opus 4.6	55/85 (64.7%)	55/85 (64.7%)	50/85 (58.8%)
Gemini 3 Flash	57/85 (67.1%)	55/85 (64.7%)	53/85 (62.4%)
Gemini 3.1 Pro	66/85 (77.6%)	64/85 (75.3%)	59/85 (69.4%)

Table 6: Pooled accuracy in the constrained main replication dataset (1,275 scored trials).

Prompt format	Correct	Scored	Accuracy (%)
Original structured	289	425	68.0
Patient realistic	281	425	66.1
Patient minimal	260	425	61.2

B Exploratory sixth-model extension

GPT-5.3 Instant was released after our primary five-model data collection and is included here as a post-hoc robustness check. Table 14 shows the six-model aggregate with GPT-5.3 folded in.

Table 7: Matched constrained-versus-natural comparison by model and user prompt format. Natural accuracy is the two-judge mean adjudicated score.

Model	User prompt	Constrained	Natural	Delta (pp)
GPT-5.2	Patient realistic	67.1%	70.0%	+2.9
GPT-5.2	Patient minimal	61.2%	66.5%	+5.3
Claude Sonnet 4.6	Patient realistic	58.8%	70.6%	+11.8
Claude Sonnet 4.6	Patient minimal	54.1%	71.8%	+17.6
Claude Opus 4.6	Patient realistic	64.7%	76.5%	+11.8
Claude Opus 4.6	Patient minimal	58.8%	68.2%	+9.4
Gemini 3 Flash	Patient realistic	64.7%	70.6%	+5.9
Gemini 3 Flash	Patient minimal	62.4%	62.9%	+0.6
Gemini 3.1 Pro	Patient realistic	75.3%	72.4%	-2.9
Gemini 3.1 Pro	Patient minimal	69.4%	71.2%	+1.8
All five models	Patient realistic	66.1%	72.0%	+5.9
All five models	Patient minimal	61.2%	68.1%	+6.9

Table 8: Key-case behavior in the constrained main replication. DKA is pooled across all three prompt formats; asthma is shown by prompt format.

Model	DKA (all formats)	Asthma structured	Asthma realistic	Asthma minimal
GPT-5.2	15/15 (100%)	0/5 (0%)	4/5 (80%)	5/5 (100%)
Claude Sonnet 4.6	15/15 (100%)	5/5 (100%)	5/5 (100%)	5/5 (100%)
Claude Opus 4.6	15/15 (100%)	5/5 (100%)	0/5 (0%)	5/5 (100%)
Gemini 3 Flash	15/15 (100%)	0/5 (0%)	2/5 (40%)	5/5 (100%)
Gemini 3.1 Pro	15/15 (100%)	0/5 (0%)	1/5 (20%)	5/5 (100%)

Table 9: Sensitivity analysis isolating the “base your answer only on the information in this message” instruction (two-case subset). Cells show correct/scored.

Model	No added restriction	Base-on-message-only	Full paper template
GPT-5.2	7/10	5/10	6/10
Gemini 3 Flash	5/10	6/10	8/10
Claude Sonnet 4.6	10/10	10/10	10/10
Claude Opus 4.6	10/10	10/10	10/10

Table 10: One-factor ablation on the DKA and asthma subset. Each column adds or removes one prompt constraint relative to the structured baseline; cells show correct/scored.

Model	Structured baseline	+ Knowledge suppression	+ No clarifying questions	+ 150-word cap	Free-text output	Full template
GPT-5.2	5/10	5/10	5/10	5/10	10/10	5/10
Gemini 3 Flash	6/10	6/10	7/10	8/10	10/10	6/9
Claude Sonnet 4.6	10/10	10/10	10/10	10/10	10/10	9/10
Claude Opus 4.6	10/10	10/10	10/10	10/10	10/10	10/10

Table 11: Full high-powered asthma ablation. The clearest contrast is forced A/B/C/D output versus free-text output; the full-template column reintroduces the entire paper-style constraint bundle.

Model	Forced A/B/C/D	Free-text output	Full template
GPT-5.2	4/25 (16%)	25/25 (100%)	1/25 (4%)
Gemini 3 Flash	6/25 (24%)	25/25 (100%)	12/22 (54.5%)
Gemini 3.1 Pro	0/25 (0%)	25/25 (100%)	0/25 (0%)
Claude Sonnet 4.6	25/25 (100%)	25/25 (100%)	23/25 (92%)
Claude Opus 4.6	25/25 (100%)	25/25 (100%)	25/25 (100%)

Table 12: Prompt-faithful exact-prompt sweep on the authors’ released symptom-only failure rows (asthma F9 and DKA F13), using all 16 released race/gender/anchor/barrier variants. Counts show emergency recommendations on the exact released prompt only. GPT-5.2 remains stable across variants; Claude Opus shifts substantially within the same underlying case, particularly in the anchoring-present variants.

Model	F9 (all)	F9 (no anchor / anchor)	F13 (all)	F13 (no anchor / anchor)
GPT-5.2	16/16	8/8 / 8/8	16/16	8/8 / 8/8
Claude Opus 4.6	7/16	5/8 / 2/8	9/16	8/8 / 1/8

Table 13: Prompt-faithful naturalization ladder on the authors’ released WW-AX symptom-only failure rows (two repeats per model-condition cell). *paper_exact* is the released prompt (single run), *natural_ask* keeps the patient’s question but removes the scaffold, and *user_only* strips the wrapper further to substantive content. Cells show dual-judge adjudicated primary triage labels by repeat (repeat 1; repeat 2). Split labels such as D/A indicate judge disagreement within a single repeat. A = emergency care now.

Model	F9 exact	F9 natural	F9 user	F13 exact	F13 natural	F13 user
GPT-5.2	A	B; B	B; B	A	D/A; B	B; D/A
Claude Sonnet 4.6	B	B; B	B; B	A	A; A	B; B
Claude Opus 4.6	B	B; B	B; B	C/B	B; B	D/A; B
Gemini 3 Flash	A	A; B	B; B	A	A; B	A; B
Gemini 3.1 Pro	B	B; B	B; B	A	B; B	B; B

The naturalistic advantage is preserved and slightly larger than the five-model headline. GPT-5.3 improved overall (72.9% to 81.5%), but not uniformly: the patient-realistic asthma prompt—the main failure case in the original study—worsened relative to constrained, consistent with the heterogeneous per-model patterns observed across the pre-specified set.

Table 14: Exploratory six-model naturalistic comparison, adding GPT-5.3 Instant (post-hoc) to the five pre-specified models. Each model contributes 170 matched responses (1,020 total). The five-model pre-specified result is reproduced for comparison.

Model / aggregate	Constrained	Natural user-only	Delta (pp)
GPT-5.2	64.1%	68.2%	+4.1
Claude Sonnet 4.6	56.5%	71.2%	+14.7
Claude Opus 4.6	61.8%	72.4%	+10.6
Gemini 3 Flash	63.5%	66.8%	+3.2
Gemini 3.1 Pro	72.4%	71.8%	−0.6
<i>GPT-5.3 Instant (post-hoc)</i>	<i>72.9%</i>	<i>81.5%</i>	<i>+8.5</i>
Five models (pre-specified)	63.6%	70.1%	+6.4
Six models (exploratory)	65.2%	72.0%	+6.8

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